Full Stack Developer Assignment: Employee Task Management System

Goal: Develop a Task Management System that showcases your abilities across frontend, backend, database design, authentication, and general problem-solving.

Core Features

1. Authentication
Support user registration, login, JWT-based authentication, a "Remember Me" option, and logout.

- Registration captures: Full Name, Email, Password, Confirm Password, and Role (Admin or Employee).
- Validation rules: email must be unique; password must be at least 8 characters and include an uppercase letter, a lowercase letter, and a number.

2. Dashboard

- *Admin view:* total employees, total tasks, completed tasks, and pending tasks.
- *Employee view:* my tasks, completed, pending, and overdue tasks.

3. Employee Management (Admin only)
Allow admins to add, edit, delete, search, sort, and paginate employees.

- Employee fields: Name, Email, Department, Designation.

4. Task Management
Enable creating, updating, deleting, and viewing tasks.

- Task fields: Title, Description, Priority, Status, Start Date, Due Date, Assigned Employee.
- Business rules:
    - Due Date must not be earlier than Start Date.
    - Completed tasks cannot be edited.
    - Employees see only their own tasks; admins see all tasks.

5. Notifications
Trigger notifications when a task is assigned, when it's due within one day, and when it's marked complete.

6. File Upload
Accept PDF, JPG, and PNG files up to 5 MB.

7. Reports
Generate Completed Tasks, Pending Tasks, and Employee-wise Task reports, with export to both Excel and CSV.

API Endpoints

- POST /register, POST /login
- Full CRUD (GET / POST / PUT / DELETE) for employees and tasks
- GET /reports

Frontend Requirements

Angular, React, TypeScript, Redux Toolkit, and React Router, with form validation and a responsive design.

Backend Requirements

Node.js + Express **or** ASP.NET Core Web API, implementing:

Bonus Features (Optional)

- Remember Me functionality
- Notifications
- File Upload (PDF/JPG/PNG, max 5 MB)
- Export to Excel/CSV
- Unit Testing
- Docker Setup
- Email Notifications